

Tourism Demand Spillovers after Hurricane Strikes

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Abstract

This study investigates the direct and spillover effects of hurricane strikes on tourist arrivals in the Caribbean. We utilize a physically based index of hurricane destruction and develop a measure of indirect hurricane exposure on nearby islands to identify spillovers. After accounting for variation in location, time, and location-specific seasonality, we find that hurricanes significantly decrease the number of tourist arrivals on the islands hit, but increase the number of arrivals on nearby islands. This tourism spillover effect is more pronounced for cruise ship arrivals than air arrivals and generally disappears after one year. This is likely due to cruise ships changing their routes to non-affected destinations in reaction to a hurricane, resulting in a significantly larger spillover. Our results suggest that policymakers could coordinate with the tourism industry to encourage the diversion of tourism by air to reduce the negative economic consequences of hurricanes in the region.

Keywords: Natural Disaster; Caribbean; Tourism Spillover; Local Projections

1 Introduction

Hurricanes significantly reduce economic activity ([Strobl 2011](#), [Deryugina et al. 2018](#)), with tourism being especially vulnerable due to its reliance on pristine natural environments and functioning infrastructure ([Becken et al. 2014](#)). Because the Caribbean is the most tourism-dependent region in the world and is particularly at risk of natural disasters ([WTTC 2022](#), [Brown et al. 2021](#)), studying the negative impact of natural hazards on tourism arrivals is crucial for the region’s economic development. For instance, hurricanes have been shown to decrease the number of arrivals in affected countries by around 2%, on average ([Granvorka & Strobl 2013](#), [Carballo Chanfón et al. 2023](#)). However, since tourism is often substitutable across countries and regions, negative shocks like hurricanes may not only impact tourism at the directly affected destination ([Rosselló et al. 2020](#)), but also have indirect consequences for travel to other, non-affected destinations ([Jin et al. 2019](#)).

Diverse shocks to tourism, including political upheavals ([Yang & Wong 2012](#)), disease outbreaks ([Groizard et al. 2022](#)), and conflict ([Karimi et al. 2022](#)), have been shown to cause tourism demand spillovers and the diversion of arrivals to other nearby countries. Given the profound impact a hurricane strike can have on local infrastructure, the perceived attractiveness of a destination, and the risk it poses to travellers, it constitutes a negative shock to both local tourism supply and demand, and thus can lead to spillovers in the tourism demand to other destinations. For cruise ship tourism, altering a route to account for a hurricane’s path is a common practice among cruise ship companies to avoid destinations that are or will be damaged by a storm ([Hochberg 2024](#), [NYTimes 2024](#), [CNN 2024](#), [Seatrade Cruise News 2024](#)). In this paper, we aim to explicitly analyze and quantify the direct impact of hurricane strikes on tourist arrivals in the Caribbean, while also considering spillover effects on nearby destinations. Accounting for spillovers is relevant in the economic assessment of natural disasters as they can be more substantial than direct effects themselves ([Hallegatte & Przyluski 2010](#)).

The region studied in this paper is the Caribbean, which is the most tourism-dependent region in the world: on average, 13.9% of GDP in 2019 (9.1% in 2021) and 15.2% of jobs (13.4% in 2021) are due to tourism ([WTTC 2022](#)), but some Caribbean territories have a tourism & travel share of GDP well above 50% and double-digit tourism growth rates. The economic importance of the Caribbean’s tourism industry is multiplied as tourism receipts surpass any other external finance source in the region and make it the

primary export sector ([Cannonier & Burke 2017](#)). In addition, tourism in the Caribbean further contributes to the economy through the multiplier effect, where increased tourism spending leads to increased overall economic growth ([Cannonier & Burke 2019](#)). The cruise ship industry is particularly concentrated in the Caribbean, accounting for around one-third of global cruise ship visits ([WTTC 2023](#)).

However, the Caribbean is also particularly exposed to the risks of natural disasters in that a significant share of the region’s assets is exposed to hurricanes and landslides, and a smaller but significant share is also at risk of floods and earthquakes ([Schweikert & Deinert 2021](#)). Between 2000 and 2019, Caribbean countries lost the equivalent of 3.6 percent of aggregate GDP per year to damages and losses related to natural hazards, which is magnitudes larger than in other developing regions ([Browne et al. 2021](#)). Hurricanes, in particular, cause devastating destruction in the region, with an estimated reduction in GDP of at least 0.83 for the average hurricane strike ([Strobl 2012](#)). Alarming, the risk of tropical storms and subsequent damages appears to have increased over the past 30 years in the region ([OCHA 2023](#)).¹ While climate change might be a driver, there has also been an increasing exposure to storms through more assets being located in risk-prone areas ([Peduzzi et al. 2012](#)).

Regional economic development in general, and tourism in particular, can be subject to both direct and spillover effects of shocks ([Faber & Gaubert 2019](#)). For natural disasters, these adjustments and spillovers can be the same or opposite sign to the initial shock and thus bias regional economic impact assessments if not explicitly taken into account ([Hallegatte & Przyluski 2010](#)). For instance, if there is a positive spillover effect of tourist arrivals to other destinations, then the hurricane strikes’ net regional economic impact is less severe. While this does not directly alleviate the local disaster and humanitarian tragedies, it opens up avenues for policies that at least compensate the affected areas for the foregone tourism revenues. One such policy is the Caribbean Catastrophe Risk Insurance Facility’s (CCRIF’s) parametric tropical cyclone insurance product, which already has had great success in practice ([CCRIF 2022](#)). This paper provides causal evidence of the tourism spillover effect following hurricanes, which can substantiate this finding and be helpful to policymakers seeking to aid a region after such shocks.

¹Hurricanes are a subtype of tropical storms that exceed certain wind speeds. The term tropical storm, therefore, includes storms caused by the same meteorological conditions that do not (yet) reach hurricane-level wind speeds.

To undertake our empirical analysis, we utilize data from the Caribbean Tourism Organization (CTO) on monthly air and cruise ship arrivals for 33 Caribbean destinations from 2000 to 2018, and link these with a hurricane destruction index based on [Strobl \(2011\)](#) and [Emanuel \(2011\)](#). To capture spillover effects, we assume that tourism spillover effects resulting from hurricane strikes are inversely proportional to the relative distance between the islands affected and other potential destinations. Since hurricane strikes have a dynamic effect on tourist arrivals ([Carballo Chanfón et al. 2023](#)), we use local projection panel regression methods to identify the effect of hurricane strikes on arrivals.

Our results show that the average hurricane strike contemporaneously reduces the number of tourists by 4.2% for arrivals by air and 37.7% by cruise ship, compared to a counterfactual baseline. At the same time, there is an average increase in arrivals of 0.7% by air and 16% by cruise ship due to spillovers. This implies that cruise tourism providers react more flexibly to hurricanes and likely adjust their routes to circumnavigate any storm. The flexible and integrated planning in the cruise ship business routing leads to a higher pass-through rate of retained arrivals compared to air travel (0.4 vs. 0.2).² In other words, a tourist traveling by cruise ship is $2\times$ more likely to substitute to another destination than a tourist traveling by air, possibly because the latter is likely to cancel the trip. Our findings suggest that there may be potential for policymakers, the tourism industry, and destinations to spread the risk of negative regional consequences through coordination.

The remainder of the paper is organized as follows. Section 2 presents the study region and discusses the literature. In Section 3 the methods and data are described, whereas Section 4 provides results. Section 5 outlines several robustness checks undertaken to support our main results. Finally, Section 6 discusses the results and concludes.

2 Literature

The empirical analysis of natural disasters in tourism finds, in general, a negative impact on the number of arrivals due to various types of natural disasters ([Rosselló et al. 2020](#)). Hurricane strikes, in particular, have been shown to decrease the number of arrivals to the affected countries and regions by around 2% on average ([Granvorka & Strobl 2013](#), [Carballo Chanfón et al. 2023](#)). Several examples in the literature provide empirical evidence

²In this case, the pass-through rate is the share of positive spillover relative to negative direct impact.

of reductions in tourist arrivals following major events. [Liu \(2014\)](#) analyze the impact of Typhoon Morakot in 2009, the deadliest typhoon to strike Taiwan in recorded history, on Taiwan’s Maolin National Scenic Area. The number of visitors to the area declined by 700’000 in the following year and a half, representing a substantial loss to the tourism business. [Mendoza et al. \(2012\)](#) investigate the impact of two earthquakes in 2007 and one in 2010 on Chile’s international tourism demand. To the best of our knowledge, however, there is no paper that specifically focuses on spillover effects in tourism due to natural disasters.

Conceptually, there are three main channels through which tourism is impacted by a hurricane such that the number of visitors decreases ([Rosselló et al. 2020](#)). First, a hurricane damages relevant infrastructure, amenities, and attractions, directly reducing the affected region’s ability to cater to tourists ([Brown et al. 2021](#)). From an economic perspective, this is the effect of a hurricane on the supply side. Second, a region affected by a hurricane might be perceived to be riskier such that potential tourists deem it unsafe ([Kozak et al. 2007](#)). Third, there might be ethical concerns for tourists traveling to a disaster region ([Becken 2015](#)).³ Together, the perceived risk and ethical concerns constitute the demand side. All three factors influence hurricanes’ direct effect on tourist arrivals.

As with direct impacts, spillover effects on tourist arrivals are also closely related to demand and supply forces, which must be carefully addressed in the empirical strategy.

⁴ When a destination cannot accommodate tourists during and following a hurricane, resulting in a decrease in supply, other destinations may benefit from the resulting surplus demand for similar forms of tourism. This mechanism represents one of the most direct and significant sources of spillover effects in tourist arrivals, and is typically observed both immediately after a hurricane strikes and in the subsequent months. Notably, this dynamic also extends to cruise ships, which must account for hurricane paths in their routing and may precautionarily bypass a destination even when the island itself is ultimately not majorly impacted ([NYTimes 2024](#), [CNN 2024](#)). In contrast to cruise tourism, which can often resume operations relatively quickly as harbors reopen, air travel-dependent tourism faces greater delays. This is due to the significant time required to repair or rebuild

³For instance, tourists might not feel unseemly to enjoy a vacation in luxury when the local population faces hardship after a disaster.

⁴For instance, [Leoni & Boto-García \(2023\)](#) take particular care in controlling for potential tourism spillovers in their analysis of the 2021 volcano eruption on La Palma, an island in the Canary archipelago (Spain), on hotel demand, supply and hospitality.

damaged accommodations and other supporting infrastructure. The disparity highlights a critical particularity of decreased supply in tourism accommodation: the reliance of air travelers on functioning land-based amenities creates a prolonged recovery period.

3 Empirical Strategy

3.1 Local Projection and Tourism Spillover

We develop a local projection panel regression model for international tourist arrivals to quantify both the effect of a direct hurricane strike and the spillover impacts of hurricane strikes on other nearby islands in the Caribbean. Local projection panel regression models were initially developed for estimating impulse response functions in multivariate dynamic systems of macroeconomic analysis (Jordà 2005). The central advantage of this approach over regular regression methods is that it directly estimates a local projection at each period of interest rather than extrapolating into increasingly distant horizons from a given model (Jordà 2023). This approach has been successfully employed in recent years to study the dynamic effect of natural disasters on economic outcomes (Naguib et al. 2022, Barattieri et al. 2023, Collalti 2024). The local projection models can be considered as a type of panel regression where, instead of including several lagged treatment indicators, a series of regressions is run on current treatment and covariates with the variable of interest shifted ahead of time. In the context of this paper, instead of asking how tourist arrivals at some time t are affected by any hurricane in the past, we ask how a hurricane at t impacted tourist arrivals m months from t :

$$\begin{aligned} \Delta Arrivals_{it+m} = & \beta_1 Hurricane_{it} + \omega_{it}(month_t \times island_i) \\ & + \delta_t + \pi_c time_t + \varepsilon_{it+m}. \end{aligned} \quad (1)$$

where the dependent variable $\Delta Arrivals_{it+m}$ is the percentage change of tourist arrivals from time t to $t + m$ on island i and $Hurricane_{it}$ is a measure of hurricane impact on island i at time t . $month_t$ and $island_i$ are month and island indicators such that their interaction includes a month-by-island fixed effect in the model, controlling for island-specific seasonality in the number of tourist arrivals. δ_t is a simple time-fixed effect that controls for shocks affecting the whole region. Lastly, $time_t$ is an island-specific time slope to account for different trajectories of tourist arrival growth across

islands, and ε_{it+m} are heteroscedasticity consistent Driscoll & Kraay (1998) standard errors with an autocorrelation length of three months. Observations are weighted by the average number of tourist arrivals in a destination so as to reduce measurement error and increase generalizability. In this setting, it is straightforward to incorporate spillovers from hurricanes on neighboring islands into the model:

$$\begin{aligned}\Delta Arrivals_{it+m} = & \beta_1 Hurricane_{it} + \beta_2 Hurricane_{it}^{other} \\ & + \beta_3 \left(Hurricane_{it} \times Hurricane_{it}^{other} \right) \\ & + \omega_{it}(month_t \times island_i) + \delta_t + \pi_c time_t + \varepsilon_{it+m}.\end{aligned}\tag{2}$$

where $Hurricane_{it}^{other}$ is a measure of hurricane impact on islands other than i , which accounts for the distance between these to capture spillovers. We not only include $Hurricane_{it}^{other}$ but also its interaction with $Hurricane_{it}$, since the two are mechanically correlated by design: if an island is hit by a hurricane and $Hurricane_{it}$ is large, then chances are high that other nearby islands are hit as well, resulting in a large $Hurricane_{it}^{other}$. In this specification, the coefficient β_1 from $Hurricane_{it}$ estimates the direct effect, whereas β_3 from the interaction $Hurricane_{it} \times Hurricane_{it}^{other}$ captures the additional spillover when other islands are affected as well.⁵

The models in Equation 1 and 2 implement a Difference-in-Differences (DiD) framework with fixed effects in a local projection setting. Three main assumptions must be fulfilled for the identification and causal interpretation of the estimates: no anticipation effect, parallel trends, and linear additive effects. In the case of the effect of a hurricane on tourist arrivals, these are, arguably, satisfied. The exact timing of a hurricane can not be forecasted before the storm starts to form. At the same time, any local seasonality in hurricane risk, which likely affects the number of tourists, is already controlled for via the fixed effects. This means that, conditional on location and season, there is a quasi-randomness in the occurrence of hurricanes that we exploit to estimate the causal effect of hurricanes on the number of tourist arrivals.

⁵The coefficient β_2 captures the spillover effect when the own island is not affected at all by a hurricane. This is almost never the case in the data, resulting in large uncertainty that cannot be interpreted. Therefore, we subsequently focus on the direct effect and the interaction spillover.

3.2 Data

3.2.1 Tourism Arrivals

Data on tourist arrivals comes from the Caribbean Tourism Organization (CTO). The CTO collects the number monthly airplane and cruise ship arrivals by destination in the Caribbean region. For our analysis we use monthly arrivals for 33 territories from 2000 to 2018 for airplane arrivals and from 2010 to 2018 for cruise ships. The list of territories included in our sample is presented in Table 1. As can be seen, these constitute mostly islands, some with independent governments, some external territories of other countries (United States, France, Netherlands, Mexico, United Kingdom). A few of the territories are not islands (Belize, Cancun, Guyana, and Suriname).⁶

3.2.2 Hurricane Destruction

To capture hurricane destruction, we follow [Strobl \(2011\)](#) in calculating the local wind exposure during a storm with the [Boose et al. \(2004\)](#) version of the [Holland \(1980\)](#) wind field model. The source of storm data used is the HURDAT Best Track Data ([Landsea & Franklin 2013](#)). Figure 1 shows the storm tracks and corresponding maximum wind speed for all storms in the study period of 2000 - 2018 that reached at least 200 km/h in their lifetime. We linearly interpolate the 6-hourly data to hourly values and run the model on an approximately 1 km $\times$ 1 km resolution of all the territories for which we have tourist arrival data. The resulting wind speed V_{kjt} is then translated to a hurricane destruction index wn_{kt} of economic impact via the non-linear damage function by [Emanuel \(2011\)](#):

$$wn_{kt} = \frac{v_{kjt}^3}{1 + v_{kjt}^3} \times 100 \quad (3)$$

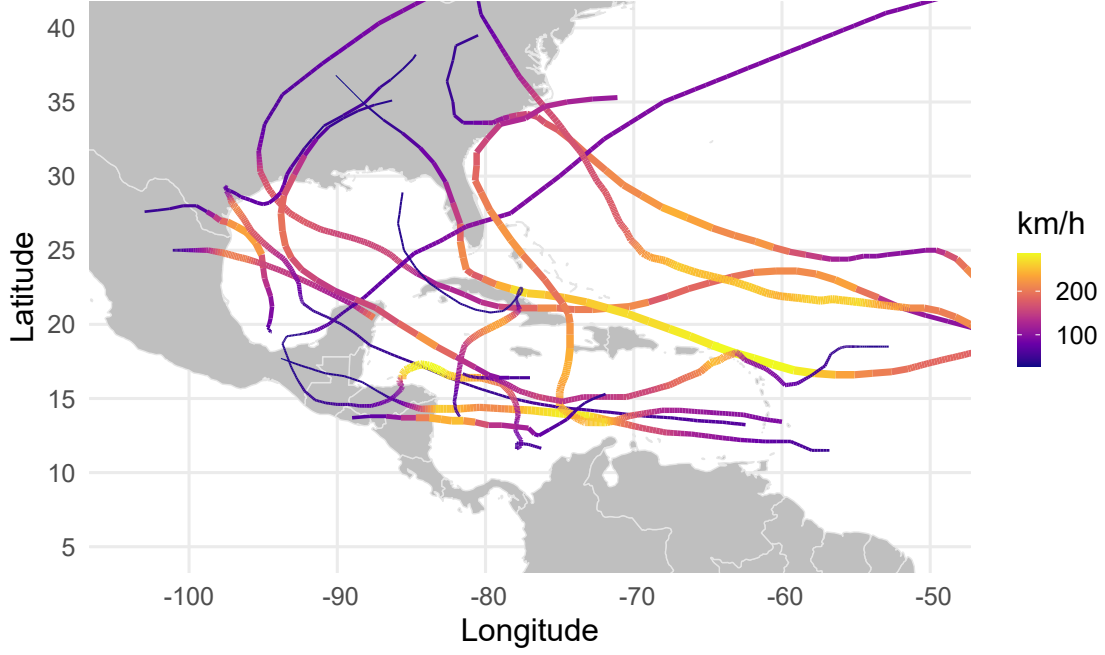
with

$$v_{kjt} = \frac{\max(V_{kjt} - V_{thresh}, 0)}{V_{half} - V_{thresh}} \quad (4)$$

where V_{kjt} corresponds to the maximum wind speed of hurricane j in location k at time t . Then, $V_{thresh} = 92$ km/h is the lower threshold below which no damages occur, whereas $V_{half} = 203$ km/h is where 50% destruction is expected. Conveniently, a one-unit increase can be interpreted as a 1% increase in damages. The maximum v_{kjt} in a given month represents the tropical cyclone impact in subsequent analysis.

⁶In the following, we refer to each territory, be it a Caribbean island or a territory on the American continent, as a destination.

Figure 1: HURDAT Storm Tracks



Notes: HURDAT storm tracks and maximum sustained wind speeds in km/h for all storms in 2000 - 2018 that reached at least 200 km/h once in their lifetime.

3.3 Destination Level Destruction and Spillover

We aggregate the hurricane destruction index on a $1 \text{ km} \times 1 \text{ km}$ grid to the destination level by considering the relative local economic development across locations. To this end, we use the distribution of night light activity from the previous year as a proxy for the relative local economic development. Specifically, we utilize a night light product that harmonizes the two primary sources of satellite night light measurements, the DMSP-OLS and the VIIRS, from 1999 to 2017 (Li & Zhou 2017, Li et al. 2020). We then calculate the island i by time t hurricane island destruction index

$$Hurricane_{it} = \sum_{k \in i} \frac{wn_{kt} \times ntl_{ky-1}}{\sum_{k \in i} ntl_{ky-1}} \quad (5)$$

where ntl_{ky-1} is the night light activity per $1 \text{ km} \times 1 \text{ km}$ cell k one year before y . In this manner, we ensure that a hurricane striking main economic centers, infrastructure, and tourist attractions is weighted more heavily in the index than when it hits remote areas.

We assume that tourist spillovers due to hurricanes are inversely proportional to the relative distance between two islands: if a hurricane strikes a nearby destination, tourism, in the form of air and cruise ship arrivals, may spill over. If the destination hit by a

hurricane is twice as far away, then the extent of spillover is reduced accordingly. More precisely, we calculate the inverse distance weighted hurricane exposure, excluding the focal destination itself, as

$$Hurricane_{it}^{other} = \sum_{j \neq i} \frac{Hurricane_{it}}{Distance_{ij}} \quad (6)$$

where $Distance_{ij}$ is the shortest distance between island i and j in km. Therefore, $Hurricane_{it}^{other}$ increases when a hurricane has hit more other territories and when these territories are closer. It follows that $Hurricane_{it}$ and $Hurricane_{it}^{other}$ are likely correlated with each other, which we account for in the local projection by including both measures and their interaction.

3.4 Summary Statistics

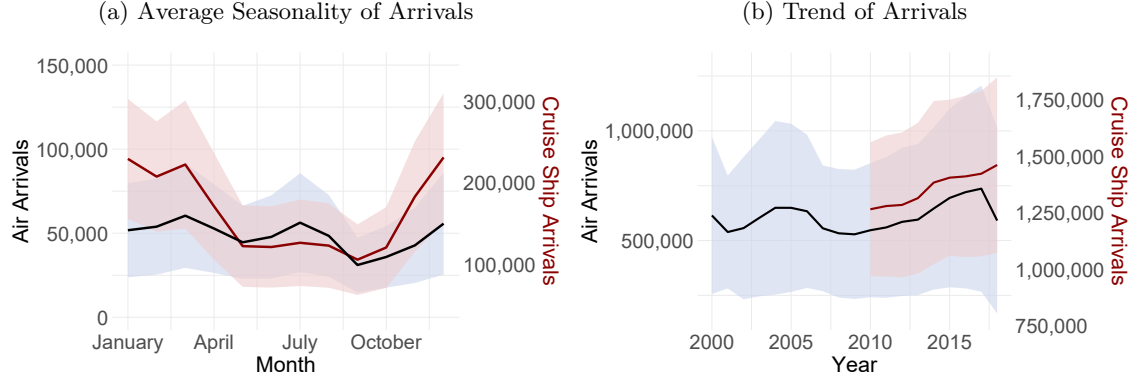
Table 1 and Table 2 in Appendix B show summary statistics of the main variables of interest per destination for airplane and cruise ship arrivals, respectively. Since data on airplane arrivals have been available since 2000 and cruise ship arrivals only since 2010, values for $Hurricane_{it}$ and $Hurricane_{it}^{other}$ differ between the two data sets.⁷ The highest numbers of average monthly airplane arrivals are to the Dominican Republic (348'400), Cuba (202'300), Jamaica (144'900), and the Bahamas (118'400). In contrast, the highest numbers of average monthly cruise ship arrivals are to the Bahamas (376'200), Cozumel (278'500), the U.S. Virgin Islands (150'500), and Sint Maarten (140'000). The worst hurricane destruction in the data was observed in Bermuda in October 2014 with Hurricanes Fay and Gonzalo (38.5% destruction), the Cayman Islands in November 2008 with Hurricane Paloma (18.3% destruction), and the Bahamas in October 2016 with Hurricane Matthew (14.5% destruction). The highest number of hurricane hits with any destruction has been observed in Cancun (24), the Bahamas (14), and Cuba (14). One may want to note that despite the large number of hurricanes that hit Cancun, only in one instance were these hurricanes damaging to other locations due to the comparatively large distance from Cancun to the other Caribbean territories.⁸ The most frequent instances of significant hurricanes on other close territories are 7 for Cuba and 6 for the Bahamas, which

⁷Note also that in Table 1, one entry is *Cancun* whereas in Table 2, it is Cozumel. That is because the main airport of the Mexican region of Quintana Roo is in Cancun, whereas the island of Cozumel is where the main cruise ship port is located.

⁸This was during Hurricane Dean 2007, when Belize was severely hit as well. Besides Hurricane Dean, Belize was not struck by any other damaging hurricane during the study period.

are both affected frequently.

Figure 2: Seasonality and Trend of Arrivals



Notes: Average number of arrivals by airplane (black) and cruise ship (red) per month (a) and year (b) across all 33 Caribbean islands. The shaded area indicates the 95% confidence interval of average monthly (a) and yearly (b) average arrivals.

Figures 2 shows the average number of air and cruise ship arrivals over time. In Subfigure 2a, one can see that the number of air arrivals peaks on average across all 33 islands in March, July, and December. September typically marks the lowest number of air arrivals. Cruise ship arrivals exhibit an even stronger seasonal pattern in terms of a high season from December to March and a low season from May to October. According to Subfigure 2b, a trend in arrivals over the years is evident. Interestingly, the number of arrivals by cruise ships has exhibited steady growth over time. In contrast, air arrivals are more volatile.⁹ For instance, there are clear signs of fewer air arrivals in 2001, 2007 - 2008, and 2018, which correspond to the years with events that caused air travel to fall: the September 11 attacks in 2001, the US subprime mortgage, and the great financial crisis of 2007 - 2008, and one of the worst hurricane seasons in late 2017.¹⁰

4 Results

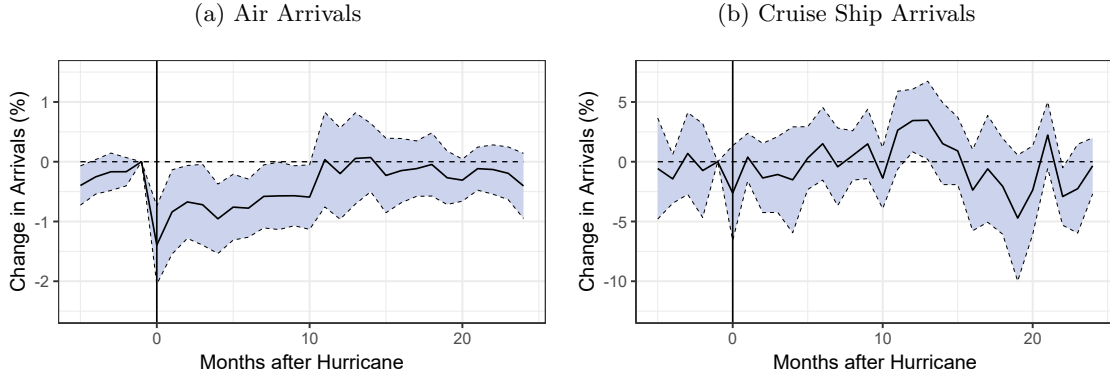
Figure 3 presents the results of estimating the local projection in Equation 1 in terms of the impulse response function for the direct impact of a hurricane. More precisely, the figure shows how the number of tourist arrivals changes per 1 percentage point increase

⁹Note that this is partly driven by the longer availability of air arrival data.

¹⁰Hurricane Maria and Irma deeply affected the Caribbean islands when they swept across the region in less than two weeks, with a total damage estimate of at least \$120B (Popke & Rhiney 2019).

in local hurricane destruction at time $t = 0$.¹¹ We see in Panel (a) that a 1% destruction reduces the number of air arrivals by around 1% to 1.5% in the months following the hurricane. The negative effect persists for several months before converging back to and stabilizing near zero within a year. One also observes no pre-trend, which supports our assumption of no anticipation.

Figure 3: Direct Effect of Hurricanes on Arrivals



Notes: Impulse response function of hurricanes on tourist arrivals from the local projection regression in Equation 1. The black line plots the coefficients with two-sided 90% confidence bands in blue. The impulse is the hurricane destruction indicator from Equation 5 such that the response can be interpreted as the percentage change in tourist arrivals per 1 percentage point higher destruction via $Hurricane_{it}$.

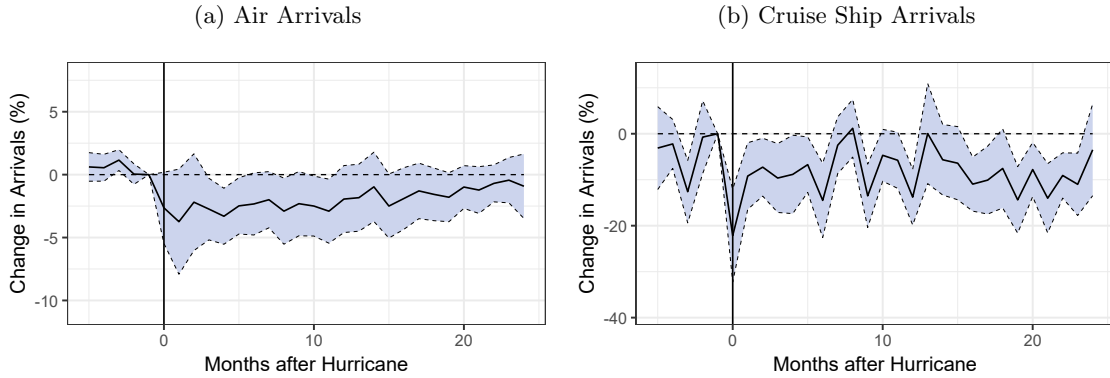
Cruise ship arrivals, as depicted in Panel (b) of Figure 3, do not exhibit a statistically significant response to hurricane events according to the model specified in Equation 1. This outcome may be attributed to one of two plausible explanations. First, hurricane strikes have no meaningful impact on cruise ship arrivals, although this interpretation appears improbable when considered alongside anecdotal evidence. Alternatively, any effects are netted out by opposing direct and spillover effects. Specifically, hurricanes may prompt cruise ships to adjust their itineraries, resulting in some destinations losing arrivals from one cruise while simultaneously gaining arrivals from rerouted cruises some days or weeks later. Consequently, on a monthly scale, the net effect on cruise ship arrivals may not diverge statistically significantly from zero. To shed some light on this, we focus on the model with spillovers in Equation 2.

Figure 4 presents the results of estimating the local projection with spillover inter-

¹¹The impulse response functions are directly available through the coefficient estimates on $Hurricane_{it}$ for different time horizons from the local projection.

action in Equation 2 in terms of the impulse response function for the direct impact of a hurricane. The impulse response function of the direct effect on air arrivals is similar to the one in Figure 3, but with a stronger negative effect of up to 3.5% per 1% in hurricane destruction and longer persistence with a recovery only after two years. Cruise ship arrivals react even more strongly than air arrivals. Per a 1 percentage point increase in the local hurricane destruction at time $t = 0$, cruise ship arrivals fall by around 22% upon impact. The negative effect then quickly reduces and stabilizes, with still consistently fewer cruise ship arrivals. Compared to air arrivals, the impulse response function of cruise ship arrivals exhibits a higher degree of variability. There are two main reasons for this. First, the panel on cruise ship arrivals is much shorter (9 years vs. 19 years), making the regression mechanically more volatile and uncertain. Second, the nature of cruise ships compared to air traffic results in more variability in the number of arrivals per month. Since cruise ships transport many tourists on set routes with weekly schedules, having one more or one less ship arriving in a month causes large variability in our data.¹²

Figure 4: Direct Effect of Hurricanes on Arrivals with Spillovers



Notes: Impulse response function of hurricanes on tourist arrivals from the local projection regression in Equation 2. The black line plots the coefficients with two-sided 90% confidence bands in blue. The impulse is the hurricane destruction indicator from Equation 5 such that the response can be interpreted as the percentage change in tourist arrivals per 1 percentage point higher destruction via $Hurricane_{it}$.

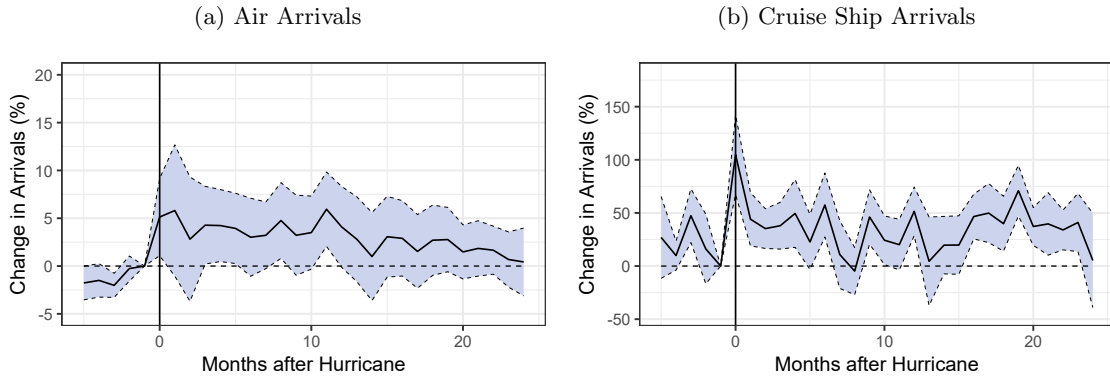
Figure 6 in Appendix C displays the spillover effect of nearby hurricanes for islands that are themselves not hit. Since our measure for a hurricane's spillover $Hurricane_{it}^{other}$ is

¹²A truly exogenous source of such variability would be the number of weekdays usually served on a specific route in a month. For instance, if a cruise ship typically arrives on Wednesday due to a certain weekly route, then having 4 or 5 Wednesdays in a month causes double-digit percentage changes in the number of cruise ship arrivals.

the distance-weighted sum of hurricane strikes on other, close, territories, there is almost no case with $Hurricane_{it} = 0$ and $Hurricane_{it}^{other} > 0$.¹³ Consequently, the impulse response function for spillovers in the form of coefficients from $Hurricane_{it}^{other}$ in Figure 6 are generally not statistically significant and have no natural use-case interpretation.

In contrast, the impulse response function of the spillover effect when territories are hit, shown in Figure 5, exhibits an interesting pattern. For air arrivals, there is a positive contemporaneous spillover effect of around 5% per one-point increase in the interaction. This effect remains positive and statistically significant and gradually converges to zero after two years. For cruise ship arrivals, this pattern is even more pronounced. Cruise ship arrivals contemporaneously increase by 100% per one-point increase in the interaction. This positive spillover effect quickly decays in the next month but stays statistically significant with the uncertainty associated with the cruise ship arrival estimates. In the following, we base our interpretation of the results on this model with the interaction term as in Equation 2.

Figure 5: Interaction Effects of Hurricanes on Arrivals



Notes: Impulse response function of hurricanes on tourist arrivals from the local projection regression in Equation 2. The black line plots the coefficients with two-sided 90% confidence bands in blue. The impulse is the interaction of the hurricane destruction indicator on the own island from Equation 5 with the hurricane destruction indicator on other islands from Equation 6. The response can, therefore, be interpreted as the percentage change in tourist arrivals per 1 percentage point higher destruction in $Hurricane_{it}^{other}$ per 1 percentage point destruction in $Hurricane_{it}$.

The effect size of the impulse response functions is best interpreted with the variation in the data in mind. The mean $Hurricane_{it}$ in the airplane data is 1.6 when there is a

¹³Econometrically, it is still prudent to include the two terms separately as well as their interaction to allow for separate and compounding effects.

damaging hurricane. Consequently, the impulse response of air arrivals due to the average hurricane is $1.6\times$ stronger than shown in Figure 4. The average hurricane, therefore, contemporaneously reduces airplane arrivals by around $1.6 \times -2.6\% = -4.2\%$. Since the mean $Hurricane_{it}$ in the cruise ship data is 1.7 when there is a hurricane, the impulse response of cruise ship arrivals in that period is even stronger. The average hurricane thus reduces cruise ship arrivals by around $1.7 \times -22.2\% = 37.7\%$ in the month of its strike.

For assessing the size of the spillover effect, we additionally need to consider $Hurricane_{it}^{other}$ conditional on $Hurricane_{it} = 0$ for the impulse response function in Figure 6 and $Hurricane_{it} > 0$ for the impulse response function Figure 5. If $Hurricane_{it} = 0$, then $Hurricane_{it}^{other}$ is close to 0 as well, implying that it is very rare not to be affected by a hurricane when other nearby territories are hit. Consequently, one should view Figure 6 as a special case of a spillover with little practical significance. On the contrary, $Hurricane_{it}^{other}$ is equal to 0.05 for the airplane data and 0.09 for the cruise ship data when $Hurricane_{it}$ is positive. Together with the average values for $Hurricane_{it}$, this means that the average value of the interaction term is $1.6 \times 0.05 = 0.08$ for the airplane and $1.7 \times 0.09 = 0.15$ for the cruise ship data. This implies that the impulse response functions in Figure 5, which do not correct for the range of $Hurricane_{it}$ and $Hurricane_{it}^{other}$, overstate the positive effect on tourist arrivals that can arise from spillovers from nearby islands due to hurricanes.

One can consider our results in terms of the contemporaneous effect of the average hurricane on air and cruise ship arrivals due to spillovers. The spillover effect for air arrivals from the interaction is an increase of $1.6 \times 0.08 \times 5.1\% \approx 0.7\%$.¹⁴ The spillover effect for cruise ship arrivals from the interaction is an increase of $1.7 \times 0.09 \times 104.9\% \approx 16\%$. Given that the average direct effect is -4.2% for air arrivals and -37.7% for cruise ship arrivals, there is still a negative net effect on average, but depending on the relative size of $Hurricane_{it}$ to $Hurricane_{it}^{other}$, a positive net effect from a hurricane strike is possible - especially for cruise ship arrivals. This would correspond to the case where other nearby territories are much more strongly affected by a hurricane strike. Comparing the relative size of negative direct and positive spillover effects for air and cruise ship arrivals, the spillover pass-through from one island to the other is about 2 times stronger for cruise ships than air arrivals.

¹⁴This is taking the coefficient estimates at $t = 0$ multiplied by the average values for $Hurricane_{it}$ and $Hurricane_{it}^{other}$ in the relevant subsample.

5 Robustness

An important consideration when comparing air and cruise ship arrival results is the significantly longer time span of air arrival data available. The additional 10 years of air arrival data, from 2000 to 2009, could partly account for the differences in the observed effects between air and cruise ship arrivals. To address this, we re-estimate the local projection model in Equation 2 for air arrivals using a restricted sample period of 2010-2018, aligning it with the timeframe available for cruise ship arrivals. Given the smaller sample size, greater estimation uncertainty is expected. Figure 7 in Appendix D presents the impulse-response functions for the restricted period (2010 - 2018) alongside those for the full period (2000 - 2018). While the general effect, significance, and shape of the curve remain similar, the range of the y-axis has been adjusted for visual display, indicating a smaller hurricane impact on air arrivals during 2010-2018 compared to 2000-2009. The main findings - that hurricanes negatively affect air arrivals and nearby hurricanes generate positive spillover effects - are robust across periods. However, the generally smaller impact of hurricanes in 2010-2018 has implications for the comparison. While cruise ship arrivals are already more impacted by hurricanes, this robustness check suggests that this is even more the case when focusing on the 2010-2018 time period.

Another simplification of our analysis is that we ignore the interplay between cruise and air arrivals. There is evidence that they are substitutes, as a drop in one can cause an increase in the other, and vice versa (Spencer et al. 2024). Addressing this interplay rigorously in a regression framework would require simultaneous equation modeling, which is beyond the scope of this study. However, it can still be insightful to view one type of arrival as a control for the other. Figure 8 in Appendix E presents impulse response functions that incorporate such controls. While the results remain broadly robust, confidence intervals are notably wider. It is important to interpret these estimates cautiously, as they involve some degree of misspecification: the two types of arrivals are jointly determined, and treating one as a control for the other is a simplistic approach that introduces potential bias. Consequently, the impulse response functions in Figure 8 primarily serve to demonstrate the general robustness of the individual effects.

6 Discussion & Conclusion

The Caribbean is particularly at risk of hurricane strikes, and its economically important tourism industry is especially vulnerable. We find that the average hurricane strike contemporaneously reduces the number of tourists by 4.2% for arrivals by air and 37.7% by cruise ship. This reduction in arrivals can be attributed to both supply-side effects, such as the destruction or damage of tourism infrastructure, amenities, and attractions, as well as demand-side effects, which make the destination appear less desirable to potential visitors. Because a considerable fraction of the Caribbean’s economy relies on tourism (WTTC 2022), such a loss in foregone tourism income is substantial, in addition to the already high reconstruction costs.

In addition to the direct negative effect, we also find a positive spillover effect on tourist arrivals following a hurricane’s strike in nearby territories. This contemporaneous positive effect is consistent with a demand spillover to another destination that serves as a substitute. Interestingly, cruise ship arrivals exhibit not only a stronger direct reduction compared to air arrivals (-37.7% vs. -4.2%) but also a stronger positive spillover effect (+16% vs. +0.7%) and a higher demand pass-through rate, on average.¹⁵ This can be explained by the nature of the cruise ship business, which is defined by set departure and arrival ports and a series of stops along the way. Cruise ships are less likely to cancel their trips and instead avoid hurricanes by adjusting their route (Zelinski 2024, NYTimes 2024). In contrast, air travelers tend to choose their destination and itinerary more independently than those on a cruise ship vacation. If a trip has to be canceled due to a hurricane, they might be less likely to reschedule their planned trip, which leads to foregone tourism revenue in the region.

Quantifying the economic impact and spillover effects of hurricanes on tourism requires understanding daily expenditures and average lengths of stay, which differ significantly between air and cruise travelers. According to the Florida-Caribbean Cruise Association (FCCA), cruise passengers spent an average of \$104 per stop-visit in the Caribbean during the 2023–2024 season (FCCA 2024), consistent with estimates from the Caribbean Tourism Organization (Carballo Chanfón et al. 2023). In contrast, stay-over tourists arriving by air spend substantially more, averaging \$1’000–\$2’000 per trip depending on the destination, with trips lasting 7–10 days. For instance, the Jamaican Tourist Board

¹⁵ $|\frac{16\%}{-37.7\%}| = 0.42$ vs. $|\frac{0.7\%}{-4.2\%}| = 0.17$

reported an average stay of 8.4 nights and daily expenditures of \$191 in 2023, amounting to approximately \$1'600 per trip (JTB 2023). Given the monthly averages of 100'000 cruise arrivals and 50'000 air arrivals per island, along with our impact estimates, we can calculate some rough economic estimates of the hurricane's impact on tourism. More specifically, an average hurricane results in 21'700 fewer cruise arrivals and 1'750 fewer air arrivals in the first month, corresponding to losses of \$2.25M and \$2.8M, respectively. If nearby islands are hit twice as hard, stronger spillover effects reduce cruise and air arrivals by only 5'700 and 1'400, respectively, resulting in losses of only \$0.59M and \$2.2M.

The presence of positive demand spillovers and the difference between the effects on air and cruise ship arrivals has several implications. For organizations assessing the regional risk of natural disasters, this implies that part of the negative economic impact of disasters on tourism is redistributed to other destinations as a surplus. From a risk and insurance perspective, this opens up the avenue of compensating affected areas by the tourism generated in nearby destinations, effectively minimizing the economic consequences of hurricanes. Increasing the demand pass-through rate and the share of arrivals that spill over to nearby destinations would also help alleviate the overall economic costs of hurricanes. Practitioners in the tourism industry could consider the significantly higher spillovers associated with cruise ship arrivals as a model to apply to other types of travel. In addition, policymakers may want to consider facilitating these measures in the tourism industry. For instance, if governments incentivized rescheduling instead of canceling air travel itineraries, similar to the cruise ship industry, and helped accommodate tourists after arrival, a substantial share of the now lost tourism demand would manifest in substitute destinations.

The main limitation of the aggregate quantitative approach in this paper is that it can not provide a microfoundation for its findings. Future research should aim to incorporate the travel decision-making process into the analysis and separately consider sources of demand and supply reduction for tourism due to hurricanes and other natural disasters. Investigating the supply effect of direct hurricane damage would entail precisely estimating the direct destruction to tourism-relevant infrastructure, amenities, and attractions. In contrast, the demand shock resulting from negative news about a hurricane could be addressed by analyzing potential visitors' news coverage and risk perception. Quantifying the other source of demand reduction, i.e., any ethical concerns of traveling to disaster regions, would require combining both qualitative and quantitative work. Another lim-

itation of this paper and a potential avenue for future research is the focus on only the Caribbean as a region and the effect of hurricanes. While the Caribbean is particularly tourism-dependent and hurricanes are a major natural disaster, making the case studied here especially relevant, extending the regional scope and disaster type is also relevant and could make the results even more applicable to policymakers in other tourism-dependent, but natural disaster-vulnerable, regions.

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A Appendix

B Data Summary Tables

Table 1: Summary Statistics: Arrivals by Airplane (2000 - 2018)

Country	<i>Arrivals</i> (in 1'000)		<i>Hurricane_{it}</i> (in %)		<i>Hurricane_{it}^{other}</i> (in %)	
	Mean	SD	Max	N	Max	N
Anguilla	5.3	2.1	4.6	4	0.7	4
Antigua & Barbuda	19.4	5.8	3.7	2	0.3	2
Aruba	68.9	21.1	0.1	1	0	0
Bahamas	118.4	34.4	14.1	14	0.4	6
Barbados	50.7	20	0	0	0	0
Belize	24.5	9.7	6.1	8	0.1	5
Bermuda	21.4	9.2	38.5	6	0.6	3
Bonaire	6.7	5.8	0	0	0	0
British Virgin Islands	26.5	11.7	0.2	1	0	1
Cancun	66.1	95.3	0.6	24	0	1
Cayman Islands	35	37.7	18.3	4	0.5	2
Cuba	202.3	105.1	2.8	14	0.1	7
Curaçao	34.9	29.2	0	1	0	0
Dominica	18.6	53.7	0	0	0	0
Dominican Republic	348.4	134.5	0.2	6	0	0
Grenada	10.3	2.8	0	0	0	0
Guadeloupe	3.3	12	0	1	0	0
Guyana	12.4	6.8	0	0	0	0
Haiti	23.1	14.6	2.5	5	0.1	1
Jamaica	144.9	50.7	1.2	5	0	1
Martinique	44.3	18.8	0	1	0	0
Montserrat	2.4	8.2	0	0	0	0
Puerto Rico	115	38.1	0.1	1	0	0
Saba	6	23.1	0	0	0	0
Sint Eustatius	1.8	4.7	0	0	0	0
Sint Maarten	35.1	13	2.2	1	0	0
St. Kitts & Nevis	8.1	3.2	0.4	1	0	1
St. Lucia	24.8	7.7	0.3	2	0	1
St. Vincent & Grenadines	8	6.7	5.3	1	0.4	1
Suriname	13.1	8.2	0	0	0	0
Trinidad & Tobago	27.5	15.4	0	0	0	0
Turks & Caicos Islands	18.2	14.6	5.3	7	0.2	2
U.S. Virgin Islands	53.8	17.8	3.6	1	0.3	1

Notes: Summary statistics of three main variables of interest per country of the airplane arrivals data.

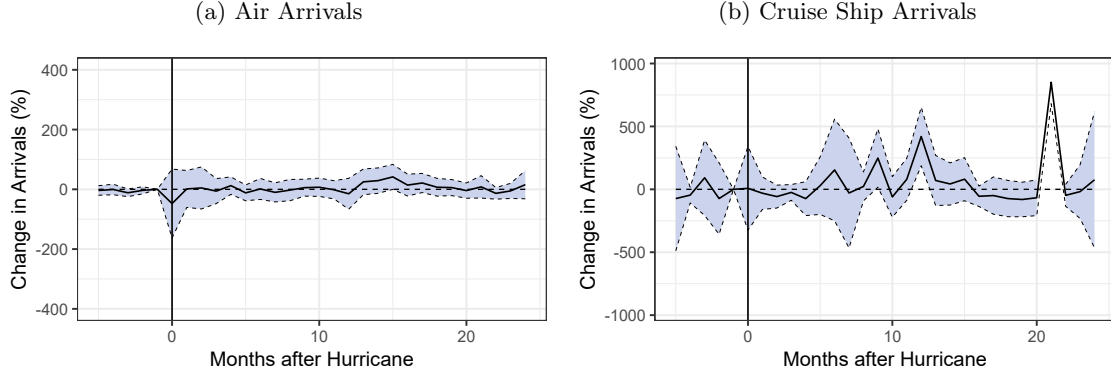
Table 2: Summary Statistics: Arrivals by Cruise Ship (2010 - 2018)

Country	<i>Arrivals</i> (in 1'000)		<i>Hurricane_{it}</i> (in %)		<i>Hurricane_{it}^{other}</i> (in ‰)	
	Mean	SD	Mean	N	Mean	N
Anguilla	7.8	5.5	4.6	3	0.7	3
Antigua & Barbuda	51.9	46.3	3.7	2	0.3	2
Aruba	55.4	33	0	0	0	0
Bahamas	376.2	70.2	14.1	5	0.4	4
Barbados	50.1	32.8	0	0	0	0
Belize	73.7	32.5	6.1	5	0.1	3
Bermuda	32.5	29.1	38.5	4	0.6	3
Bonaire	14.8	16	0	0	0	0
British Virgin Islands	36.4	34.2	0.2	1	0	1
Cayman Islands	134.9	47.2	0	0	0	0
Cozumel	278.5	84	0	0	0	0
Cuba	0	0	2.8	5	0.1	2
Curaçao	44.7	26.2	0	0	0	0
Dominica	23.1	24.9	0	0	0	0
Dominican Republic	49.3	42.3	0.2	3	0	0
Grenada	23.7	25	0	0	0	0
Guadeloupe	17.8	24.5	0	1	0	0
Guyana	0	0	0	0	0	0
Haiti	54.4	20.5	2.5	3	0.1	1
Jamaica	120.7	48.8	1.2	1	0	1
Martinique	16.7	24.3	0	1	0	0
Montserrat	0.2	0.3	0	0	0	0
Puerto Rico	107.9	44.4	0.1	1	0	0
Saba	0	0	0	0	0	0
Sint Eustatius	0	0	0	0	0	0
Sint Maarten	140	71.5	2.2	1	0	0
St. Kitts & Nevis	62.1	49.4	0.4	1	0	1
St. Lucia	53.7	37	0.3	2	0	1
St. Vincent & Grenadines	9.4	12.2	5.3	1	0.4	1
Suriname	0	0	0	0	0	0
Trinidad & Tobago	6	8.2	0	0	0	0
Turks & Caicos Islands	67.9	21.4	5.3	4	0.2	1
U.S. Virgin Islands	150.5	65.3	3.6	1	0.3	1

Notes: Summary statistics of three main variables of interest per country of the cruise ship arrivals data.

C Direct Spillover Effects

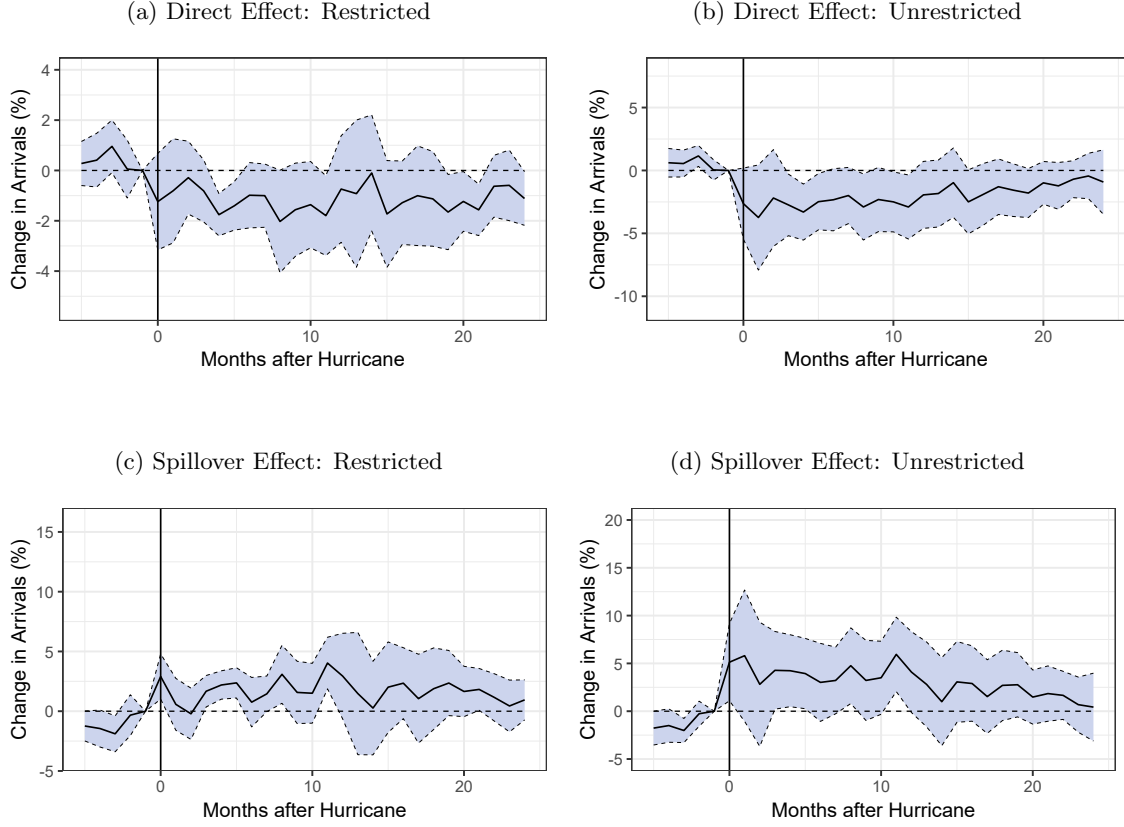
Figure 6: Spillover Effects of Hurricanes on Arrivals



Notes: Impulse response function of hurricanes on tourist arrivals from the local projection regression in Equation 2. The black line plots the coefficients with two-sided 90% confidence bands in blue. The impulse is the hurricane destruction indicator on other islands from Equation 6 such that the response can be interpreted as the percentage change in tourist arrivals per 1 percentage point higher destruction in $Hurricane_{it}^{other}$.

D Restricted vs. Unrestricted Sample

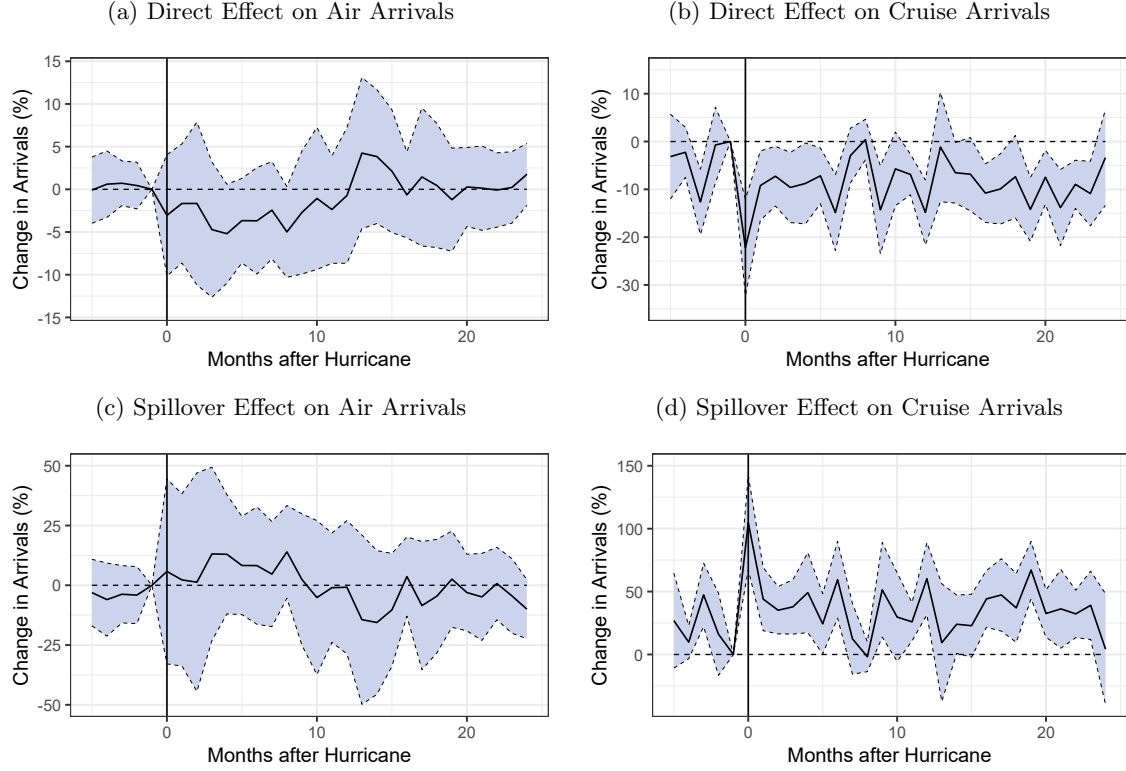
Figure 7: Effect of Hurricanes on Air Arrivals: Restricted vs. Unrestricted Sample



Notes: Impulse response function of hurricanes on tourist arrivals from the local projection regression in Equation 2. The black line plots the coefficients with two-sided 90% confidence bands in blue. Impulse response functions with unrestricted data correspond to the use of air arrival data in the full period 2000 - 2018, whereas restricted data correspond to the period 2010 - 2018 with the availability of cruise ship data.

E Substitution between Arrival Types

Figure 8: Effect of Hurricanes on Air Arrivals with Substitution between Arrival Type



Notes: Impulse response function of hurricanes on tourist arrivals from the local projection regression in Equation 2 but additionally controlling for cruise ship arrivals in the estimation for air arrivals and vice versa. The black line plots the coefficients with two-sided 90% confidence bands in blue.